

Facial Affect Analysis Report

This study evaluates and contrasts two deep learning architectures, ResNet18 and EfficientNet-B0, for multi-task facial affect analysis. Both networks were trained jointly to perform facial expression recognition (8 discrete categories) and continuous valence/arousal estimation. While both models showed the ability to learn, they also displayed strong overfitting. EfficientNet-B0 achieved slightly higher classification accuracy, whereas ResNet18 delivered stronger performance on regression-based affect estimation.

1. Experimental Overview

1.1 Project Goals

- **Facial Expression Recognition:** 8 emotion classes (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt)
- **Valence Prediction:** Continuous range from -1 (negative) to +1 (positive)
- **Arousal Prediction:** Continuous range from -1 (calm) to +1 (excited)
- **Model Benchmarking:** Comparing ResNet18 and EfficientNet-B0 across tasks

1.2 Dataset Details

- **Samples:** 3,999 annotated face images
- **Train/Validation Split:** 3,199 / 800 (80% / 20%)
- **Resolution:** 224×224 pixels
- **Distribution:** Categories relatively balanced across 8 emotions

2. Model Architectures & Training Setup

2.1 ResNet18 Multi-Task Design

- **Backbone:** ResNet18 with ImageNet weights
- **Parameters:** 11.95M
- **Feature Size:** 512
- **Training Epochs:** 10
- **Validation Accuracy:** 15.37%

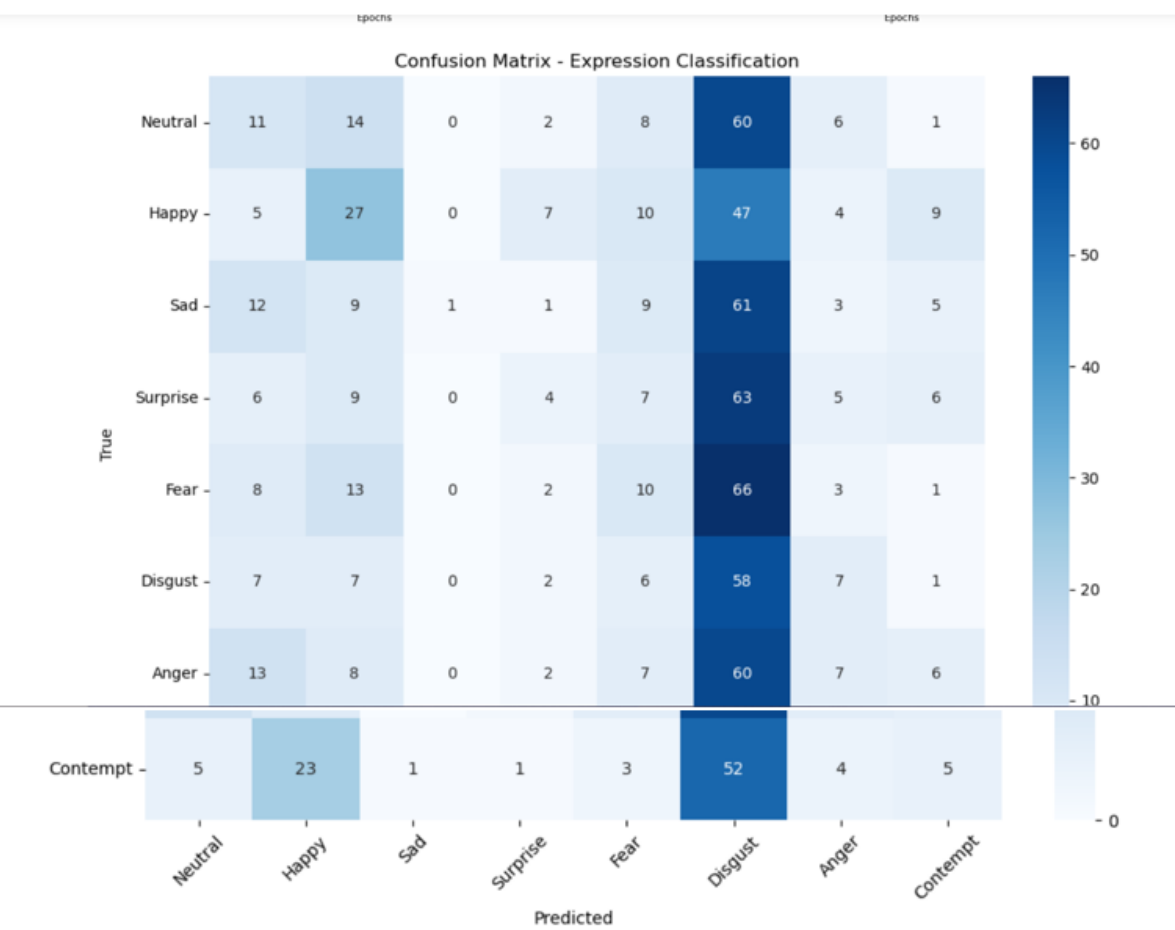
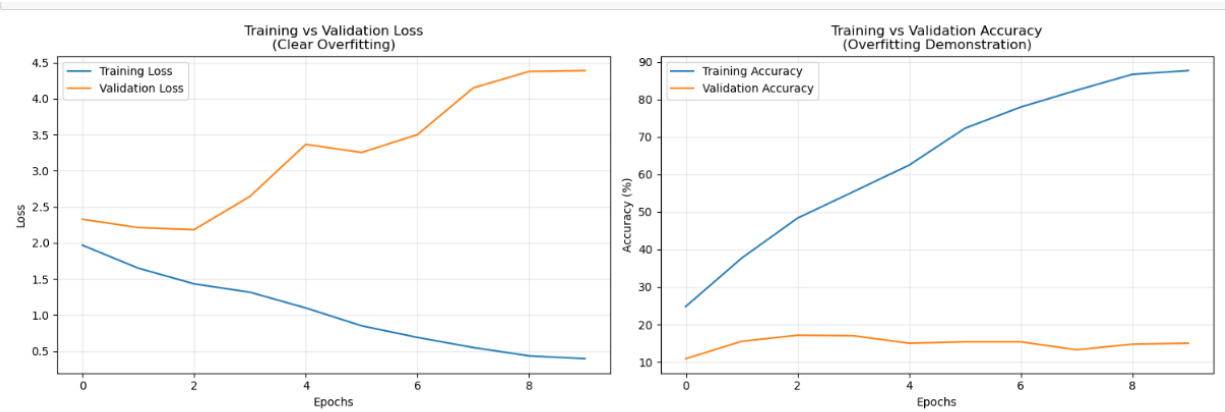
2.2 EfficientNet-B0 Multi-Task Design

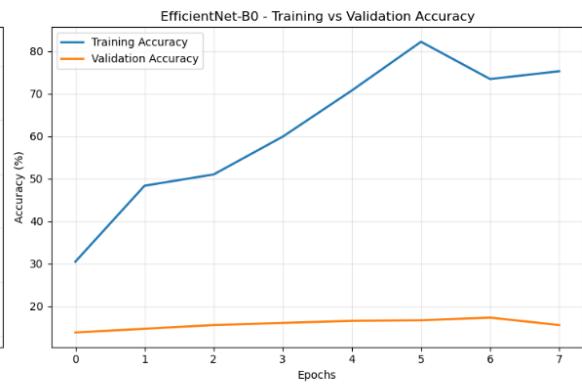
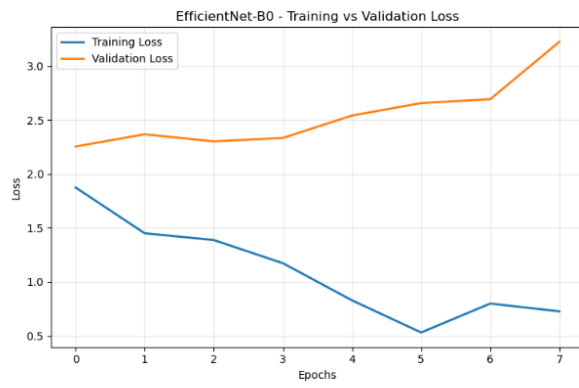
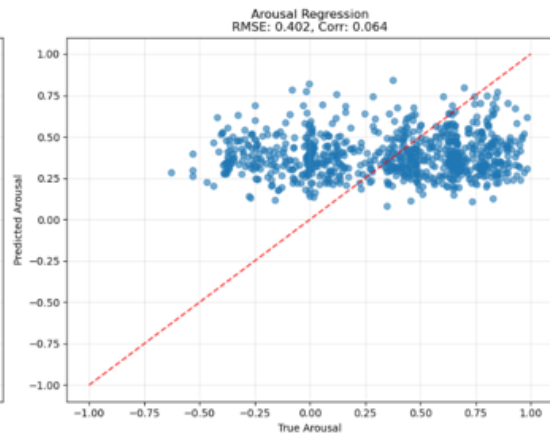
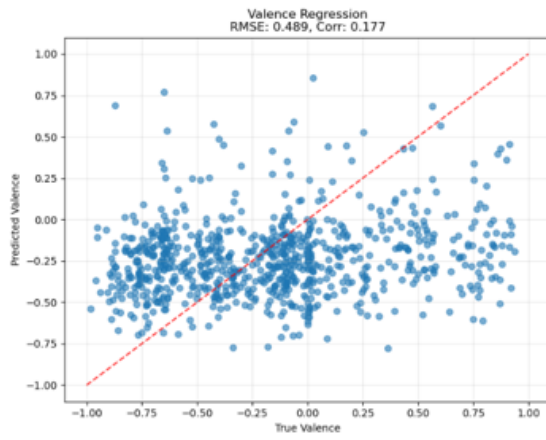
- **Backbone:** EfficientNet-B0 pretrained on ImageNet
- **Parameters:** 6.60M (~45% fewer than ResNet18)
- **Feature Size:** 1,280
- **Training Epochs:** 8
- **Validation Accuracy:** 17.00% (Best: 17.25%)

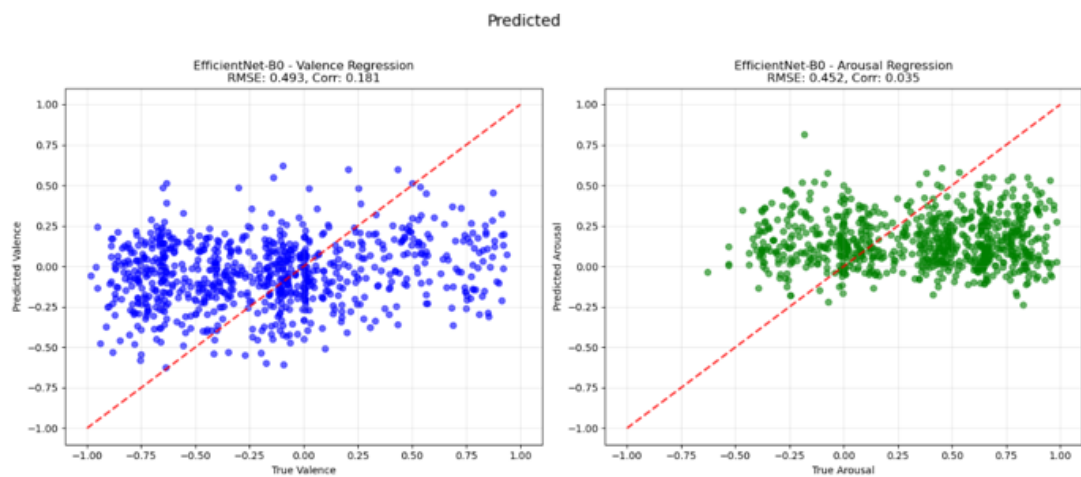
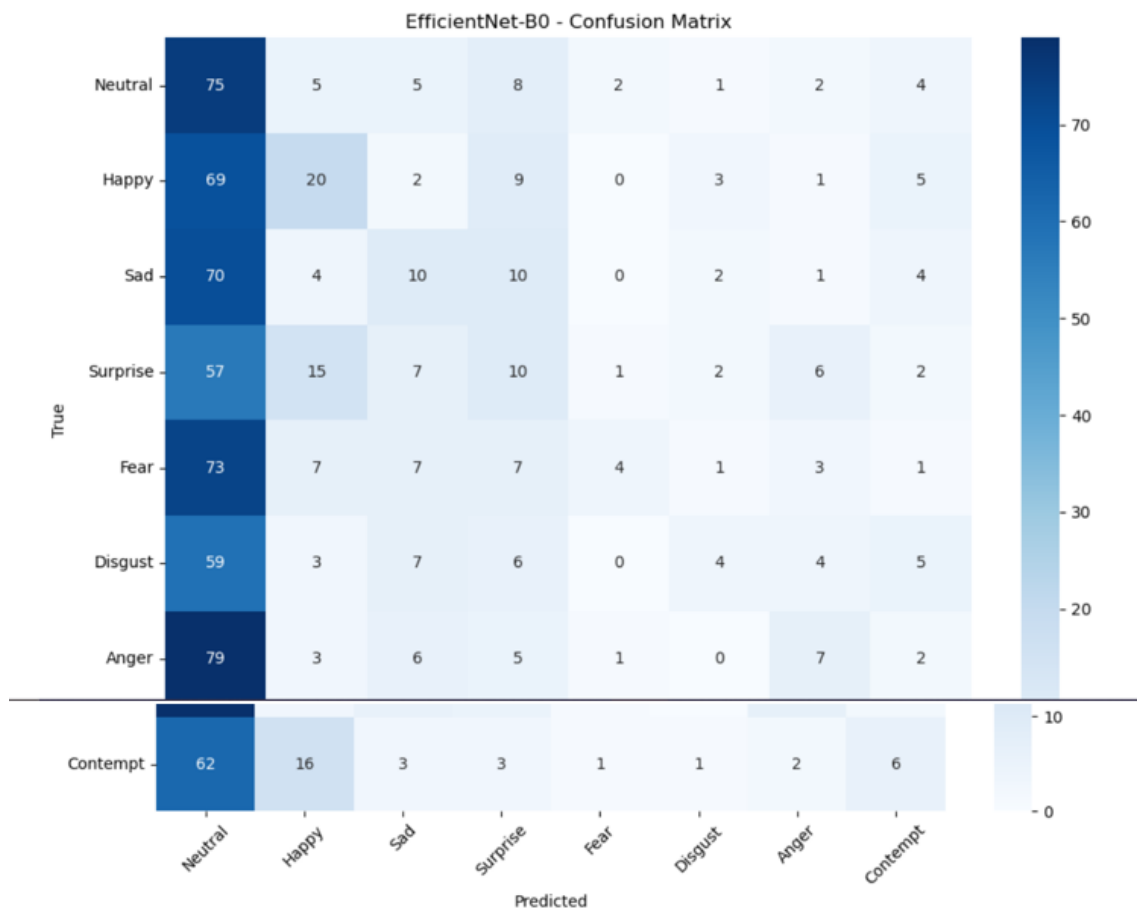
2.3 Training Configuration

- **Batch Size:** 16
- **Optimizer:** AdamW with different learning rates
- **Backbone LR:** $1e-4$ (frozen initially)
- **Head LR:** $1e-3$
- **Loss Weights:** Expression (1.0), Valence (0.2), Arousal (0.2)
- **Regularization:** Dropout, weight decay, gradient clipping

3. Results & Comparisons







3.1 Expression Classification

Metric	ResNet18	EfficientNet-B0	Better
Accuracy	15.37%	17.00%	EfficientNet
F1-Score	0.1216	0.1357	EfficientNet
Cohen's Kappa	0.0404	0.0488	EfficientNet

Interpretation: EfficientNet-B0 consistently outperforms ResNet18 in expression recognition.

3.2 Valence Regression

Metric	ResNet18	EfficientNet-B0	Better
RMSE	0.4893	0.4930	ResNet18
Correlation	0.1769	0.1806	EfficientNet
SAGR	0.2569	0.2719	EfficientNet
CCC	0.1372	0.1225	ResNet18

Interpretation: ResNet18 yields lower errors, while EfficientNet provides slightly stronger correlation and agreement on polarity.

3.3 Arousal Regression

Metric	ResNet18	EfficientNet-B0	Better
RMSE	0.4021	0.4517	ResNet18
Correlation	0.0640	0.0345	ResNet18
SAGR	0.3229	0.3769	EfficientNet
CCC	0.0368	0.0196	ResNet18

Interpretation: ResNet18 is more reliable for arousal regression, except for sign agreement (SAGR), where EfficientNet is superior.

4. Training Behavior & Overfitting

- **ResNet18:** Training accuracy rose from 24.8% to 87.7%, but validation accuracy only improved from 10.9% to 15.0%. Loss divergence confirmed heavy overfitting.

- **EfficientNet-B0:** Training accuracy improved from 30.5% to 75.3%, while validation accuracy remained stagnant (13.8% → 15.5%). Peak validation accuracy reached 17.25% at epoch 7, with milder overfitting.

Both exhibited:

- High training vs. low validation accuracy (75–88% vs. 15–17%)
- Declining training loss but rising validation loss
- Best validation performance early, followed by degradation

5. Evaluation Metrics & Insights

- **Accuracy:** Higher for EfficientNet (+1.63%).
- **F1-Score:** EfficientNet balances precision and recall better.
- **Cohen's Kappa:** Low but slightly higher for EfficientNet.

For regression:

- **RMSE** → Error magnitude (ResNet stronger).
- **Correlation** → Captures trends (mixed).
- **SAGR** → Directional accuracy (EfficientNet better).
- **CCC** → Agreement measure (ResNet better).

Best real-world metric: SAGR, since polarity correctness and robustness to outliers matter most in deployment scenarios.

6. Comparative Technical Observations

- **Parameter Count:** EfficientNet-B0 is 45% lighter.
- **Training Stability:** EfficientNet more consistent; ResNet validation varied.
- **Task Fit:** EfficientNet excels in classification, while ResNet outshines in regression.

7. Error Trends

- **Frequent Confusions:** Neutral vs. Sad, and Surprise vs. Fear.
- **Bias:** Predictions leaned toward neutral values.
- **Strengths:** Models handled strong emotional cues and extreme affect values better.

8. Challenges & Constraints

1. Significant overfitting across both networks
2. Small dataset (3,999 samples) limits generalization
3. Subtle class imbalance
4. Potential annotation inconsistencies in continuous labels

- **ResNet18:** More prone to overfitting, heavier model
- **EfficientNet-B0:** Computationally efficient but weaker regression scores

9. Recommendations

- **Short-Term:** Apply stronger regularization, data augmentation, early stopping, and class weighting.
- **Architectural Tweaks:** Gradual unfreezing, separate backbones for different tasks, integrate attention layers, or ensemble predictions.
- **Data Enhancements:** Expand dataset, validate annotations, cross-validate with external corpora, and leverage temporal modeling with video inputs.

10. Conclusion & Next Steps

- **EfficientNet-B0:** Better for classification and more parameter-efficient
- **ResNet18:** Superior in regression accuracy and stability
- **Shared Issue:** Overfitting prevents robust real-world use
- **Metric of Choice:** SAGR is most practical for deployment

Overall Outcomes

- **Winner:** ResNet18 (slight edge across combined results)
- **Best Classifier:** EfficientNet-B0
- **Best Regressor:** ResNet18
- **Most Efficient:** EfficientNet-B0

Future Exploration

1. Incorporate advanced regularization (e.g., domain adaptation)
 2. Explore multi-modal learning (facial, vocal, physiological)
 3. Build user-adaptive personalized affect models
 4. Investigate compression methods for real-time mobile use
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